

Computer Technology Integration and Student Learning: Barriers and Promise

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Published online: 20 September 2008
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Abstract Political and institutional support has enabled many institutions of learning to spend millions of dollars to acquire educational computing tools (Ficklen and Muscara, *Am Educ* 25(3):22–29, 2001) that have not been effectively integrated into the curriculum. While access to educational technology tools has remarkably improved in most schools, there is still concern about instructional integration of computer technology to support student learning. Appropriate integration of computer tools constitutes a major change in people's lives; technology integration is a complex phenomenon that involves understanding teachers' motivations, perceptions, and beliefs about learning and technology (Woodbridge, <http://www.techlearning.com/story/showArticle.php?articleID=17701367>). Although computer technology has a great potential to reform or even transform education, barriers come in the way of achieving success especially with student learning. This paper presents an overview of technology use in education with a focus on barriers to computer technology integration. Further, this paper provides suggestions to maximize the benefits of learning with computer technology.

Keywords Educational technology · Computer tools · Instructional integration · Student learning

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Introduction

The history of using computer technology for learning is replete with promises and disappointments (Rosenberg 2001); technology has not fulfilled its potential to bring significant and desirable changes in education (Ginserb and McCormick 1998). Further, it is mostly underutilized, and has not been implemented in very effective or creative ways. Despite reports of increased use of instructional technology on many campuses (Green 2001; Jacobsen 2000), majority of teachers do not integrate computer technology into their curriculum. Cuban (2001) contends that computers, in most institutions, are most often employed to supplement traditional classroom pedagogy and are yet to be effectively integrated into daily classroom practices.

While educators have witnessed increased access to the Internet, greater bandwidth, and computer tools including mobile technologies, many questions remain unanswered as to whether or not the large financial investments in technology is impacting student learning (Burnham et al. 2000). Contrary to the lack of evidence to show that increased technological capital has improved education at all (Cuban 2001; Oppenheimer 2003; Rogers 2000), many educators, government, and business leaders still believe that computer investment enhances computer instructional use and improves teaching and learning. Even so, neither providing computer tools in the classroom (Candiotti and Clark 1998) nor providing state-of-art technology by itself will make any desirable learning changes in education (Kent and McNergney 1999).

Political and institutional support has enabled many institutions to spend millions of dollars to acquire educational computing tools (Ficklen and Muscara 2001), which, regrettably, have not been effectively integrated into the

curriculum. Institutions that spend more money in technology resources argue that technological expansion lowers costs associated with maintaining physical facilities of an institution (Debard and Guidera 2000). This practice, however, does not necessarily enhance effective technology integration, but it continues to put more pressure on teachers who are reportedly not integrating technology into their courses.

A report by the National Center for Education Statistics (2000) indicated that 44% of teachers used technology for classroom instruction, 42% for computer applications, 12% for practice drills, 41% required research using the Internet, 27% had students conduct research using CD-ROMs, 27% assigned multimedia projects, 23% assigned graphical presentations of materials, 21% assigned demonstrations, 20% required students to use technology to solve problems and analyze data, and 7% assigned students to correspond with others using the Internet. These low figures imply that effective integration of computer technology into classroom instruction has yet to be realized.

Many teachers use technology less today than they did in the mid-1980s; surveys of teachers show consistent declines in the classroom use of technologies. Cuban et al. (2001) reported that fewer than 20% of teachers use technology several times a week, and up to half of all teachers did not use technology at all. Cuban (2001) identifies various reasons for this sad trend, including lack of time to learn how to use the technology and develop specific computer activities for classroom use. Cuban (2001) notes:

It is premature to call the investment in computers in schools a failure because of lack of evidence for increased productivity and transformed teaching and learning. As the infrastructure matures and teachers' beliefs about teaching and learning evolve, more and more teachers will change their practices and become serious users of computers in their university and public classrooms (p. 179).

Computer literacy previously emphasized in teacher education programs does not seem to work. A technology brief from the American Association of Colleges for Teacher Education (AACTE) highlights this plight:

Our ultimate goal has not yet been achieved. It is not enough for professors to use technology in their office, or even to use presentation software in their classroom. We must design courses that require our students to use technology themselves-only then will they be prepared to incorporate technology into the lessons they will teach their own students (Wetzel 2001 p. 5).

Technology has pervaded all sectors of education prompting the need to prepare teachers to take advantage of

these tools. Although technologies allow students to work more productively than in the past, the teacher's role in technology-rich classrooms is more demanding than ever. According to the International Society for Technology in Education (2000):

Today's classroom teachers must be prepared to provide technology-supported learning opportunities for their students...being prepared to use technology and knowing how that technology can support student learning must be integral skills in every teacher's professional repertoire (p. 2).

Teachers are constantly challenged to prepare graduates who can effectively use technology as a learning tool, yet the teachers themselves are new to various technology uses and have no personal experiences as students themselves in technology-infused classrooms (Jacobsen et al. 2002). To be successful teaching with technology requires teachers to have a strong comfort level with, and consistently implement technology tools as part of their own repertoire of tools in courses they are teaching. Additionally, research has shown that, for instance, that teacher modeling technology use in teaching methods courses is the best means for preparing pre-service teachers to integrate technology into their own classroom teaching (Zehr 1997).

Barriers to Computer Technology Integration

Teachers who may be committed to integrating computer technology in the classroom may find the process challenging due to the barriers that exist. The barriers are categorized as external (first-order) or internal (second-order). According to Snoeyink and Ertmer (2001), first-order barriers include lack of equipment, unreliability of equipment, lack of technical support and other resource-related issues. Second-order barriers include both school-level factors, such as organizational culture and teacher-level factors, such as beliefs about teaching and technology and openness to change.

Fabry and Higgs (1997) identified resistance to change, negative attitudes toward computers, constraints on training and support, cost, and a lack of access to the right types of technology in appropriate locations as key barriers to integration of technology in the classroom. However, according to Dias (1999), one often overlooked yet strong barrier to integration of educational technology in the classroom is change. Harris and Sullivan (2000) points out that:

When teachers are asked to integrate technology they are really being asked to change in two ways. First, they are asked to adopt new teaching tools such as the

computer and the Internet. These are vastly different tools from the classroom tools many currently use such as the chalkboard, overhead projector, or television. Second, teachers are asked to change the way they teach their students, which may include changing the role they play in the classroom and the way their classrooms are physically arranged (p. 12).

Various studies (Beggs 2000; Newhouse 1999) reported lack of time, lack of equipment, and lack of training as barriers to technology integration. Lack of time is also a big barrier: teachers need time to learn how to use the hardware and software, time to plan, and time to collaborate with other teachers. Teachers also need time to develop and incorporate technology into their curriculum. Some teachers are unable to make appropriate use of technology in their own classrooms, while others are unwilling to try because of anxiety, lack of interest, or lack of motivation (Duhaney 2001). Most teachers cite ineffective technological leadership in their institutions as a big challenge to computer technology integration (Duhaney 2001; Krueger et al. 2000). Cuban (2001) reported lack of time and poor technical support as barriers to technology integration at technology-rich Stanford University. Effective teacher need ample time to effectively organize and prepare instructional materials (Stronge 2002).

In summary, major barriers identified by teachers in the use and integration of computer technology in the classroom include: lack of computers and relevant quality software, lack of time, lack of funding, technical problems, teacher attitude toward computers, lack of teacher confidence, resistance to change, poor administrative support, poor training, and a lack of vision to integrate technology into the curriculum. However, the real challenges of integrating information technology into the classroom depend on the behavior, investments, and commitment of individual teachers.

Computer Technology Integration and Student Learning

Computer technology integration is lacking throughout the education system (NCES 2000). Integrating technology in the curriculum requires knowledge of the subject area, an understanding of how students learn and a level of technical expertise (Morgan 1996). This implies that teachers need to be comfortable with computers in order to use or integrate them in their courses (Milone 1999; Wang 2000). Computer technology integration is likely to occur in learning environments that are supported by active, constructive, collaborative, intentional, conversational, contextualized, and reflective learning (Jonassen 2000).

Further, true technology integration is centered on a more meaningful use of technology in schools (ISTE 2000).

To enhance computer use and integration in education, Sprague et al. (1998) suggest various tenets of effective teacher model that include: (a) awareness of what the technology can offer; (b) opportunity to explore technology integration; (c) time to learn the technology; (d) application of technology to teaching; and, (e) reflection on teaching. The above tenets, along with improved access, Sprague et al. (1998) argue, provide a sound basis for strong professional development opportunities and positive models for teachers to learn how to effectively integrate computer technology into their instruction.

While it is deemed essential to have computers in the classrooms to ensure true integration (Maddux 1998), access is one thing and integration is another. To integrate computers into the classroom, teachers should feel confident and comfortable using computers through the use of computers on a consistent basis for instructional activities. Teachers must understand the value of computing in education to be able to benefit their students and to support meaningful learning (Novak 1998). For instance, teachers who understand and support computer-based constructivist methodologies are likely to develop the expertise to incorporate computers into their instruction, integrate computers technology into their courses, and help their students to do the same.

Bitner and Bitner (2002) suggest that a teacher's attitude and skills determine the effectiveness of technology integration in the curriculum. However, while training teachers to integrate technology into instruction in a seamless and transparent way (Dvorak and Buchanan 2002), technology training frequently produces less than desirable effects for a number of reasons including lack of time, funds, and direct connection of the technology training to the curriculum (Rakes and Casey 2002). Further, teachers need to gain some technology skills but they will be most successful using technology as a learning tool for their students when they do not act as experts but as guides (Jonassen et al. 1999).

Integration of technology can be achieved when students learn with computers in ways that include: using computers efficiently and effectively in the general content areas to allow students to learn how to apply computer skills in a meaningful way; using real life software applications that will help students learn to use computers flexibly, purposefully and creatively; having the curriculum drive technology as opposed to technology driving the curriculum; and, organizing the goals of curriculum and technology into a coordinated, harmonious whole (T.H.E Journal 1999). Some element of creativity is also required of teachers if the use of technology is to have a positive impact on student learning (Mehlinger and Powers 2002).

Teachers generally teach the way they were taught (Mehlinger and Powers 2002) and infusing technological tools into instruction poses unique challenges to instructors who are not ready and willing to change to modern constructivist teaching styles. Many teachers lack role models to guide them through the necessary changes they will need to make to be successful in integrating new technology into their classroom (Johnson and Liu 2000). Existing diffusion theories emphasize how important such leaders are for acceptance within the larger social system (Rogers 1995). Consequently, it is important for education leaders to provide leadership with a model of technology integration that can produce feasible results.

To ensure effective integration of new technologies in education requires commitment from administrators; administrators should strive to provide in-depth professional development to the faculty and other staff. Components of professional development for effective technological use should include: (a) connection to student learning; (b) hands-on technology use; (c) variety of learning experiences; (d) curriculum specific applications; (e) new role for teachers; (f) collegial learning; (g) active participation of teachers; (h) on-going process; (i) sufficient time, (j) technical assistance and support; (k) administrative support; (m) adequate resources; (n) continuous funding; and, (o) built-in evaluations (Rodriguez and Knuth 2000).

Some teachers wonder whether and how computer-based activities address educational objectives, especially when they lack a clear vision for appropriate technology use (Fuller 2000). Further, “teachers refuse to incorporate computer use into their practice because they feel threatened by the values it embodies” (p. 512). Further, teaching with technology requires extensive investments of time that exerts a lot of demand on most teachers (Loeding and Wynn 1999). As a result, teachers may resist using technology as it takes time away from their actual instruction and other responsibilities that they care more about.

In line with constructivist perspectives, teachers using technology in the classroom should allow students to explore areas of technology that they are unfamiliar with themselves, thus empowering the students to learn with technology (Brooks and Brooks 2001). The constructivist approach is based on the understanding that students learn more when they take responsibility for their own learning (Henson 2004). Further, the instructor assumes a new role of guiding, mediation, modeling, and coaching learners (Sharp 2006). Anderson and Becker (2001) argue that “constructivist-oriented teachers use computers in more varied ways, have greater technical expertise in the use of computers, use computers frequently with students, and use them in more powerful ways” (p. 55).

To be successful in computer use and integration, teachers need “to engage in conceptual change regarding their beliefs about the nature of learning, the role of the student, and their role as teacher” (Niederhauser et al. 1999, p. 157). Instructors who are successful in teaching with technology strive constantly to learn better ways to infuse technology into their courses in order to help students to effectively learn with technology (Anderson and Becker 2001). Further, “to move beyond the integration level in the technology integration hierarchy, the teacher must be able to relinquish that control to the students” (Nisan-Nelson 2001, p. 91).

Conclusion

Technology presents educators with challenges and opportunities to educate students to their highest potential. However, providing technology resources without effectively integrating them into instruction will not produce better learners (Tolmie 2001). Schools should strive to create a strong vision grounded on technology integration and offer relevant professional development programs that support teachers experimenting with new educational technologies. Dede (2000) argues that unless simultaneous innovations occur in pedagogy, assessment, and other primary areas of education reform, technology by itself would not help improve education but instead continue to “reinforce many educators’ cynicism about fads based on magical machines. (p. 185)

Teachers should strive to do accomplish the following objectives: (a) Create relationships between active learning and active teaching; (b) Develop an appreciation and an understanding of the potential of technology; (c) Develop leadership skills and become role models for successful integration; (d) Understand the power of technology adoption and integration; (e) Design integrated curriculum activities; (f) Learn the benefits of technology in the classroom; (g) Develop ownership of the technology through authentic experiences; (h) Learn to motivate students with technology; (i) Become advocates for technology integration and usage.

According to Jacobsen et al. (2002), the real challenge is to “develop fluency with teaching and learning with technology, not just with technology, itself” (p. 44). Rather than viewing technology as merely a tool for delivery, it should be seen as a means to improve learning. Jacobsen et al., observe, “When you begin to think differently about technology and learning and you have different spaces in which to learn and teach, you can design different approaches to learning” (p. 27).

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